

The Beveridge Response to a Destruction Shock under Infinitely Elastic Vacancy Creation

Notes for review (not for the draft)

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Numbers are produced by `Programs baseline/gs_free_entry_beveridge.jl` (output `gs_free_entry_beveridge` block “Exposition tables”). The program checks every closed form below by assertion against a simulated impulse response of the same linear system.

1 Setting

We work with the model under the conditions of the draft’s independence proposition (`prop:independence`): $\sigma = 0$, $\zeta = 0$ and $p_0 = 0$. The labor-market block then closes on its own. It is the Gabrovski–Silva economy with a time-varying destruction rate δ_t . In this block:

- the discount factor is β , and $r \equiv 1/\beta - 1$ is the discount rate;
- recruiter compensation $y \equiv z/\mu$ is constant;
- exit is exogenous, so a business survives from t to $t + 1$ with probability $1 - \delta_t$;
- every vacated position is reposted at no cost.

For exposition we set the per-match cost $\kappa = 0$, so all recruiting costs are sunk ($x_v = 1$). Remark 3 discusses $\kappa > 0$.

Timing We use the draft’s convention. δ_t is known at the start of t and destroys businesses at the end of t . Hence u_t and the inherited vacancy stock are fixed at t . Tightness θ_t and entry e_t respond at t . The destroyed jobs and positions appear in u_{t+1} and v_{t+1} .

Matching $q(\theta) = A\theta^{-\eta_L}$ and $f(\theta) = \theta q(\theta) = A\theta^{1-\eta_L}$, with $\eta_L \in (0, 1)$.

2 The nonlinear system

2.1 The vacancy value and its flow value under free entry

New vacancies require a sunk cost drawn from $[0, x_m]$. With infinitely elastic creation, the value of a vacancy is pinned whenever entry is positive:

$$Q_t = x_m \quad \text{for all } t. \tag{1}$$

The flow value of holding a vacancy for one period (eq:Kdef in the draft) is $K_t = Q_t - \beta(1 - \delta_t)Q_{t+1}$. The firm holds an asset worth Q_t , and it is worth Q_{t+1} next period only if the business survives destruction at the end of t . Substituting (1),

$$K_t = x_m[1 - \beta(1 - \delta_t)] = x_m(1 - \beta + \beta\delta_t). \quad (2)$$

Q is a constant. K_t is a known function of the exogenous rate δ_t , not an endogenous variable. When δ is constant, as in Gabrovski–Silva, K is constant too. When δ_t moves, K_t moves with it, because a vacancy is an asset and δ_t is its depreciation rate. Holding K fixed instead would give the flow-cost DMP model (Remark 2). We substitute (2) wherever K appears, so K is not carried as a variable.

2.2 The remaining equations

$$\begin{aligned} \frac{x_m(1 - \beta + \beta\delta_t)}{q(\theta_t)} &= \beta(1 - \delta_t) \mathbb{E}_t \left[(1 - \phi)(y - b - x_m(1 - \beta + \beta\delta_{t+1})) \right. \\ &\quad \left. + (1 - s) \frac{x_m(1 - \beta + \beta\delta_{t+1})}{q(\theta_{t+1})} - \phi \theta_{t+1} x_m(1 - \beta + \beta\delta_{t+1}) \right], \end{aligned} \quad (3)$$

$$u_{t+1} = u_t - (1 - \delta_t)f(\theta_t)u_t + [\delta_t + s(1 - \delta_t)](1 - u_t), \quad (4)$$

$$v_{t+1} = (1 - \delta_t)[(1 - q(\theta_t))v_t + s(1 - u_t)] + e_{t+1}, \quad (5)$$

$$\theta_t = v_t/u_t. \quad (6)$$

(3) Job creation

- *The left side is the expected cost of a hire.* The vacancy's flow value $x_m(1 - \beta + \beta\delta_t)$ is paid for an expected $1/q(\theta_t)$ periods.
- *The right side is the value of a match.* The match first produces at $t + 1$, and only if the business survives.
- *The bracket is the firm's surplus after substituting the Nash wage.* It has three terms:
 - The first is the firm's share of output, net of the worker's outside option and of the vacancy's flow value, which a matched firm forgoes.
 - The second is the hiring cost saved next period because the match continues.
 - The third is the part of that saving the worker captures through the wage. The worker's outside option includes finding another job, and $\phi fK/q = \phi\theta K$ because $f(\theta)/q(\theta) = \theta$.

(4) Unemployment Next period's unemployed are:

- today's unemployed, less those hired at a business that survives;
- plus the employed who lose their job, to destruction (δ_t) or to separation at a surviving business ($s(1 - \delta_t)$).

(5) Vacancies Next period's vacancies are:

- today's unfilled vacancies that survive destruction;
- positions vacated by separations at surviving businesses and reposted at no cost;
- new entry.

How free entry orders the system Equation (3) contains only θ and the exogenous δ . Given the path of δ , it determines the path of θ on its own. Then:

- (4) gives u ;
- (6) gives $v = \theta u$;
- (5) only determines entry e_{t+1} , as whatever delivers $v_{t+1} = \theta_{t+1} u_{t+1}$ given the inherited stock.

3 Steady state

Write

$$\bar{K} \equiv x_m(1 - \beta + \beta\bar{\delta}) = x_m \frac{r + \bar{\delta}}{1 + r}, \quad \bar{\tau} \equiv \bar{\delta} + \bar{s}(1 - \bar{\delta}).$$

$\bar{K}$ is the steady-state flow value of a vacancy. It is a user cost: the asset price times the discount rate plus the destruction rate. The steady-state versions of (3) and (4) are

$$\frac{\bar{K}}{\bar{q}} = \beta(1 - \bar{\delta}) \left[(1 - \phi)(y - b - \bar{K}) + (1 - \bar{s}) \frac{\bar{K}}{\bar{q}} - \phi \bar{\theta} \bar{K} \right], \quad (7)$$

$$\bar{u} = \frac{\bar{\tau}}{\bar{\tau} + (1 - \bar{\delta})\bar{f}}, \quad \text{equivalently} \quad \frac{1 - \bar{u}}{\bar{u}} = \frac{(1 - \bar{\delta})\bar{f}}{\bar{\tau}}. \quad (8)$$

4 Log-linear system

Hats denote log deviations: $\hat{x}_t \equiv \ln x_t - \ln \bar{x}$. For the destruction rate we use the level deviation $d_t \equiv \delta_t - \bar{\delta}$.

4.1 Job creation

Take logs of both sides of (3).

Left side

$$\ln \frac{x_m(1 - \beta + \beta\delta_t)}{q(\theta_t)} = \ln x_m + \ln(1 - \beta + \beta\delta_t) - \ln A + \eta_L \ln \theta_t.$$

Its first-order deviation is

$$\frac{\beta}{1 - \beta + \beta\bar{\delta}} d_t + \eta_L \hat{\theta}_t = \frac{d_t}{r + \bar{\delta}} + \eta_L \hat{\theta}_t,$$

using $\beta/(1 - \beta + \beta\bar{\delta}) = 1/(r + \bar{\delta})$.

Right side The log of $\beta(1 - \delta_t)$ deviates by $-d_t/(1 - \bar{\delta})$. Write Σ_{t+1} for the bracket in (3). Its steady-state value $\bar{\Sigma}$ satisfies $\bar{K}/\bar{q} = \beta(1 - \bar{\delta})\bar{\Sigma}$ by (7). The deviation of Σ_{t+1} is the sum of three terms:

$$\begin{aligned} d \left[(1 - \phi)(y - b - x_m(1 - \beta + \beta\delta_{t+1})) \right] &= -(1 - \phi) \beta x_m d_{t+1} = -(1 - \phi) \bar{K} \frac{d_{t+1}}{r + \bar{\delta}}, \\ d \left[(1 - s) \frac{x_m(1 - \beta + \beta\delta_{t+1})}{q(\theta_{t+1})} \right] &= (1 - \bar{s}) \frac{\bar{K}}{\bar{q}} \left(\frac{d_{t+1}}{r + \bar{\delta}} + \eta_L \hat{\theta}_{t+1} \right), \\ d \left[-\phi \theta_{t+1} x_m(1 - \beta + \beta\delta_{t+1}) \right] &= -\phi \bar{\theta} \bar{K} \left(\frac{d_{t+1}}{r + \bar{\delta}} + \hat{\theta}_{t+1} \right). \end{aligned}$$

Divide the sum by $\bar{\Sigma} = \bar{K}/[\bar{q}\beta(1-\bar{\delta})]$, and use $\bar{\theta}\bar{q} = \bar{f}$:

$$\hat{\Sigma}_{t+1} = \beta(1-\bar{\delta}) \left[-(1-\phi)\bar{q} \frac{d_{t+1}}{r+\bar{\delta}} + (1-\bar{s}) \left(\frac{d_{t+1}}{r+\bar{\delta}} + \eta_L \hat{\theta}_{t+1} \right) - \phi \bar{f} \left(\frac{d_{t+1}}{r+\bar{\delta}} + \hat{\theta}_{t+1} \right) \right].$$

Equating the two sides Set the left-side deviation equal to $-d_t/(1-\bar{\delta}) + \mathbb{E}_t \hat{\Sigma}_{t+1}$. Write $B \equiv \beta(1-\bar{\delta})$, move the d_t term to the right, and collect terms:

$$\boxed{\eta_L \hat{\theta}_t = - \left[\frac{1}{r+\bar{\delta}} + \frac{1}{1-\bar{\delta}} \right] d_t + B \mathbb{E}_t \left\{ [1-\bar{s} - \phi \bar{f} - (1-\phi)\bar{q}] \frac{d_{t+1}}{r+\bar{\delta}} + [(1-\bar{s})\eta_L - \phi \bar{f}] \hat{\theta}_{t+1} \right\}} \quad (9)$$

Only $\hat{\theta}$ and the shock appear.

Interpretation of each weight

- η_L on $\hat{\theta}_t$. A tighter market lengthens the time to fill, so a hire costs more.
- $1/(r+\bar{\delta})$ on d_t : *the rental effect*. A one-point rise in destruction raises the vacancy's flow value $x_m(1-\beta+\beta\delta_t)$ by βx_m , which is $1/(r+\bar{\delta})$ in proportion. The longer-lived the vacancy, the larger the proportional rise. This raises the cost of a hire.
- $1/(1-\bar{\delta})$ on d_t : *the survival effect*. A hire pays off only if the business survives. This lowers the value of a hire. It is about 1.
- On d_{t+1} , which is nonzero only if the shock persists:
 - $1-\bar{s}-\phi\bar{f}$: a higher flow value next period raises the value of an ongoing match, which saves next period's hiring cost, net of the worker's share;
 - $-(1-\phi)\bar{q}$: a matched firm forgoes that flow value, so the firm's surplus falls.
- On $\hat{\theta}_{t+1}$:
 - $(1-\bar{s})\eta_L$: a tighter market next period raises the hiring cost an ongoing match saves;
 - $-\phi\bar{f}$: it also raises the worker's outside option and hence the wage.

4.2 Unemployment

Write the right side of (4) as $U(u_t, \delta_t, \theta_t) = u_t - (1-\delta_t)A\theta_t^{1-\eta_L}u_t + [\delta_t + s(1-\delta_t)](1-u_t)$. Its partial derivatives at the steady state are

$$\begin{aligned} \frac{\partial U}{\partial u_t} &= 1 - (1-\bar{\delta})\bar{f} - [\bar{\delta} + \bar{s}(1-\bar{\delta})] = 1 - (1-\bar{\delta})\bar{f} - \bar{\tau}, \\ \frac{\partial U}{\partial \delta_t} &= \bar{f}\bar{u} + (1-\bar{s})(1-\bar{u}), \\ \frac{\partial U}{\partial \theta_t} &= -(1-\bar{\delta})\bar{u}(1-\eta_L)\frac{\bar{f}}{\bar{\theta}}. \end{aligned}$$

Hence

$$u_{t+1} - \bar{u} = [1 - (1-\bar{\delta})\bar{f} - \bar{\tau}](u_t - \bar{u}) + [\bar{f}\bar{u} + (1-\bar{s})(1-\bar{u})]d_t - (1-\bar{\delta})\bar{u}(1-\eta_L)\bar{f}\hat{\theta}_t.$$

Divide by $\bar{u}$. The coefficient on d_t becomes, by (8),

$$\frac{\bar{f}\bar{u} + (1 - \bar{s})(1 - \bar{u})}{\bar{u}} = \bar{f} + (1 - \bar{s})\frac{(1 - \bar{\delta})\bar{f}}{\bar{\tau}} = \frac{\bar{f}}{\bar{\tau}}[\bar{\tau} + (1 - \bar{s})(1 - \bar{\delta})] = \frac{\bar{f}}{\bar{\tau}},$$

because $(1 - \bar{s})(1 - \bar{\delta}) = 1 - \bar{\tau}$. Define

$$\lambda \equiv 1 - (1 - \bar{\delta})\bar{f} - \bar{\tau}, \quad \omega \equiv (1 - \bar{\delta})\bar{f}(1 - \eta_L).$$

Then

$$\boxed{\hat{u}_{t+1} = \lambda \hat{u}_t + \frac{\bar{f}}{\bar{\tau}} d_t - \omega \hat{\theta}_t} \quad (10)$$

Interpretation of each weight

- λ is *persistence*. An extra unemployed worker is still unemployed next month unless hired, with probability $1 - (1 - \bar{\delta})\bar{f}$. The $\bar{\tau}$ term reflects that one fewer employed worker means one fewer separation.
- $\bar{f}/\bar{\tau}$ is the *destruction weight*. A one-point rise in δ_t adds two groups to unemployment: the employed who lose their job, $(1 - \bar{s})(1 - \bar{u})$, and the new matches lost before producing, $\bar{f}\bar{u}$. Relative to the stock $\bar{u}$ this is $\bar{f}/\bar{\tau}$. It is large because unemployment is a small stock, of order $\bar{\tau}$. It depends on $\bar{f}$ and $\bar{\tau}$, not on how $\bar{\tau}$ splits into δ and s .
- ω is the *hiring channel*. Lower tightness at t means fewer hires at t , who are still unemployed at $t + 1$.

4.3 Vacancies

Write the right side of (5) as $V(v_t, u_t, \delta_t, \theta_t, e_{t+1}) = (1 - \delta_t)[(1 - A\theta_t^{-\eta_L})v_t + s(1 - u_t)] + e_{t+1}$. Its partial derivatives at the steady state are

$$\begin{aligned} \frac{\partial V}{\partial \delta_t} &= -[(1 - \bar{q})\bar{v} + \bar{s}(1 - \bar{u})] \equiv -\bar{X}, & \frac{\partial V}{\partial v_t} &= (1 - \bar{\delta})(1 - \bar{q}), \\ \frac{\partial V}{\partial \theta_t} &= (1 - \bar{\delta})\eta_L \frac{\bar{q}}{\bar{\theta}} \bar{v}, & \frac{\partial V}{\partial u_t} &= -(1 - \bar{\delta})\bar{s}, & \frac{\partial V}{\partial e_{t+1}} &= 1. \end{aligned}$$

Hence

$$v_{t+1} - \bar{v} = -\bar{X}d_t + (1 - \bar{\delta})(1 - \bar{q})(v_t - \bar{v}) + (1 - \bar{\delta})\eta_L \bar{q} \bar{v} \hat{\theta}_t - (1 - \bar{\delta})\bar{s}(u_t - \bar{u}) + (e_{t+1} - \bar{e}).$$

Divide by $\bar{v}$, and use $\bar{u}/\bar{v} = 1/\bar{\theta}$:

$$\boxed{\hat{v}_{t+1} = -\frac{\bar{X}}{\bar{v}} d_t + (1 - \bar{\delta})(1 - \bar{q}) \hat{v}_t + (1 - \bar{\delta})\eta_L \bar{q} \hat{\theta}_t - (1 - \bar{\delta})\frac{\bar{s}}{\bar{\theta}} \hat{u}_t + \frac{\bar{e}}{\bar{v}} \hat{e}_{t+1}} \quad (11)$$

Interpretation of each weight

- $\bar{X}/\bar{v}$. This is the share of next period's stock inherited from this period. Destruction removes it one for one.
- $(1 - \bar{\delta})(1 - \bar{q})$. This is the share of vacancies that are neither filled nor destroyed.

- $(1 - \bar{\delta})\eta_L\bar{q}$. A tighter market fills fewer vacancies, so more carry over. After a destruction shock $\hat{\theta} < 0$, so vacancies fill faster.
- $(1 - \bar{\delta})\bar{s}/\bar{\theta}$. Reposts come from the employed, who are fewer when u is higher.
- $\bar{e}/\bar{v}$. This is entry's share of the stock.

At free entry, (11) determines $\hat{e}_{t+1}$, not $\hat{v}_{t+1}$.

4.4 Tightness

Taking logs of (6),

$$\hat{v}_t = \hat{\theta}_t + \hat{u}_t \quad (12)$$

4.5 Weights at the calibration

Our calibration holds $\bar{f}$, $\bar{q}$ and $\bar{\tau}$ fixed and recalibrates $\bar{s}$ and ϕ at each $\bar{\delta}$.

	D1 ($\bar{\delta} = 3.2\%/yr$)	CK-like ($\bar{\delta} = \bar{\tau}$)
$1/(r + \bar{\delta})$	167.2	29.2
$\bar{f}/\bar{\tau}$	13.26	13.65
λ	0.559	0.559
ω	0.164	0.164
$\bar{X}/\bar{v}$	0.930	0.174
$\bar{e}/\bar{v}$	0.073	0.831

Table 1: Weights of the log-linear system.

The unemployment equation is essentially the same in both economies. Among the weights that matter for u and v at free entry, only $1/(r + \bar{\delta})$ differs much.

5 Tightness

Lemma 1. *Let $d_t = \rho^t d_0$ with $\rho \in [0, 1)$, and suppose*

$$\Phi \equiv B \left[(1 - \bar{s}) - \frac{\phi \bar{f}}{\eta_L} \right] > -1.$$

Then (9) has a unique bounded solution, $\hat{\theta}_t = a d_t$, with

$$a = - \frac{\frac{1}{1 - \bar{\delta}} + \frac{1 - B\rho[1 - \bar{s} - \phi\bar{f} - (1 - \phi)\bar{q}]}{r + \bar{\delta}}}{\eta_L - B\rho[(1 - \bar{s})\eta_L - \phi\bar{f}]} < 0. \quad (13)$$

Proof. Uniqueness. Divide (9) by η_L :

$$\hat{\theta}_t = \Phi \mathbb{E}_t \hat{\theta}_{t+1} - \frac{1}{\eta_L} \left[\frac{1}{r + \bar{\delta}} + \frac{1}{1 - \bar{\delta}} \right] d_t + \frac{B[1 - \bar{s} - \phi\bar{f} - (1 - \phi)\bar{q}]}{\eta_L(r + \bar{\delta})} \mathbb{E}_t d_{t+1}.$$

Here $\Phi < B(1 - \bar{s}) < 1$, and $\Phi > -1$ by assumption. So the unique bounded solution is obtained by solving forward. With $\mathbb{E}_t d_{t+k} = \rho^k d_t$, every term of the forward sum is proportional to d_t , so $\hat{\theta}_t = a d_t$ for some constant a .

The coefficient. Substitute $\hat{\theta}_t = a \rho^t d_0$, $\mathbb{E}_t \hat{\theta}_{t+1} = a \rho^{t+1} d_0$, $d_t = \rho^t d_0$ and $\mathbb{E}_t d_{t+1} = \rho^{t+1} d_0$ into (9):

$$\eta_L a \rho^t d_0 = - \left[\frac{1}{r + \bar{\delta}} + \frac{1}{1 - \bar{\delta}} \right] \rho^t d_0 + B \left\{ [1 - \bar{s} - \phi \bar{f} - (1 - \phi) \bar{q}] \frac{\rho^{t+1} d_0}{r + \bar{\delta}} + [(1 - \bar{s}) \eta_L - \phi \bar{f}] a \rho^{t+1} d_0 \right\}.$$

Divide by $\rho^t d_0$:

$$\eta_L a = - \frac{1}{r + \bar{\delta}} - \frac{1}{1 - \bar{\delta}} + \frac{B \rho [1 - \bar{s} - \phi \bar{f} - (1 - \phi) \bar{q}]}{r + \bar{\delta}} + B \rho [(1 - \bar{s}) \eta_L - \phi \bar{f}] a.$$

Collect the terms in a on the left:

$$a \left\{ \eta_L - B \rho [(1 - \bar{s}) \eta_L - \phi \bar{f}] \right\} = - \frac{1}{1 - \bar{\delta}} - \frac{1 - B \rho [1 - \bar{s} - \phi \bar{f} - (1 - \phi) \bar{q}]}{r + \bar{\delta}},$$

which is (13).

The sign. $B \rho < 1$ and $1 - \bar{s} - \phi \bar{f} - (1 - \phi) \bar{q} < 1$, so the numerator of the second fraction is positive. The denominator satisfies $\eta_L - B \rho [(1 - \bar{s}) \eta_L - \phi \bar{f}] \geq \eta_L [1 - B \rho (1 - \bar{s})] > 0$. Hence $a < 0$. $\square$

Remark 1 (Why the endogenous states do not enter). The economy has endogenous states, u_t and the predetermined vacancies $v_{pre,t}$. A general linear solution for tightness would take the form $\hat{\theta}_t = a d_t + b_u \hat{u}_t + b_v \hat{v}_{pre,t}$. The proof above nevertheless delivers $b_u = b_v = 0$, for the following reasons.

1. Equation (9) contains neither u nor v_{pre} . Free entry pins Q , and $K_t = x_m(1 - \beta + \beta \delta_t)$ depends only on the exogenous δ_t .
2. Iterating (9) forward T periods gives

$$\hat{\theta}_t = \sum_{k=0}^{T-1} \Phi^k \mathbb{E}_t (c_0 d_{t+k} + c_1 d_{t+k+1}) + \Phi^T \mathbb{E}_t \hat{\theta}_{t+T},$$

where c_0 and c_1 are the coefficients in the proof. For a bounded solution with $|\Phi| < 1$, the last term vanishes as $T \rightarrow \infty$. So $\hat{\theta}_t$ depends only on current and expected future d , which gives $b_u = b_v = 0$.

3. The states are then solved backward. Iterating (10) gives $\hat{u}$ (Proposition 1, Part 2), and (12) gives $\hat{v} = \hat{\theta} + \hat{u}$.
4. The predetermined stock $v_{pre,t+1} = (1 - \delta_t)[(1 - q(\theta_t))\theta_t u_t + s(1 - u_t)]$ depends only on $(u_t, \theta_t, \delta_t)$ and has no own lag. It feeds into nothing except entry, $e_{t+1} = v_{t+1} - v_{pre,t+1}$.

Root count. The system has one unstable root, $1/\Phi$, for one jump variable, and one stable root, λ , for one dynamic state, u . So the Blanchard–Kahn condition holds.

When the structure breaks. With finite ξ , entry $e = G(Q)$ links v_{pre} to Q , and Q enters the job creation condition. Tightness then depends on the states, the guess must include state coefficients, and the closed form is lost. The same happens when the conditions of **prop:independence** fail, through N , C or χ^c .

Reading (13)

- $1/(1 - \bar{\delta})$ is the survival effect.
- $1/(r + \bar{\delta})$ is the rental effect.
- Persistence adds two offsetting adjustments, both because the flow value stays high next period:
 - it raises the value of an ongoing match, through $1 - \bar{s} - \phi \bar{f}$;
 - it erodes next period's surplus, through $-(1 - \phi)\bar{q}$.
- The denominator accounts for tightness being low next period as well.
- At $\rho = 0$, $a = -[1/(r + \bar{\delta}) + 1/(1 - \bar{\delta})]/\eta_L$.

6 Proposition

Proposition 1 (Destruction shock under infinitely elastic vacancy creation). *Assume the setting of Section 1, with $\xi \rightarrow \infty$, the condition $\Phi > -1$ of Lemma 1, and $\lambda \in [0, 1)$. Let $\delta_t = \bar{\delta} + \rho^t d_0$ with $d_0 > 0$ and $\rho \in [0, 1)$, and let $a < 0$ be given by (13). To first order:*

1. Impact. $\hat{u}_0 = 0$ and $\hat{v}_0 = \hat{\theta}_0 = a d_0 < 0$.

2. Unemployment. For $t \geq 1$,

$$\hat{u}_t = \left(\frac{\bar{f}}{\bar{\tau}} + \omega|a| \right) d_0 g_t > 0, \quad g_t \equiv \sum_{j=0}^{t-1} \lambda^{t-1-j} \rho^j.$$

3. Vacancies at $h = 1$. $\hat{v}_1 < 0$ if and only if

$$|a|(\rho - \omega) > \frac{\bar{f}}{\bar{\tau}}. \quad (14)$$

4. Vacancies at later horizons. If $\hat{v}_t \geq 0$ for some $t \geq 1$, then $\hat{v}_{t'} > 0$ for all $t' > t$. Moreover, $\hat{v}_t < 0$ for every $t \geq 0$ if and only if

$$\rho > \lambda \quad \text{and} \quad |a|(\rho - \lambda - \omega) \geq \frac{\bar{f}}{\bar{\tau}}.$$

5. The whole path. The slope of a regression of $\tilde{v}_t \equiv v_t - \bar{v}$ on $\tilde{u}_t \equiv u_t - \bar{u}$ along the impulse response ($t \geq 0$) is

$$\frac{\sum_t \tilde{v}_t \tilde{u}_t}{\sum_t \tilde{u}_t^2} = \bar{\theta} \left[1 - \frac{|a| \rho (1 - \lambda^2)}{(\bar{f}/\bar{\tau} + \omega|a|)(1 + \rho\lambda)} \right]. \quad (15)$$

It is negative if and only if

$$|a|[\rho(1 - \lambda^2) - \omega(1 + \rho\lambda)] > \frac{\bar{f}}{\bar{\tau}}(1 + \rho\lambda). \quad (16)$$

Corollary 1 (iid shock). *If $\rho = 0$, then $\hat{\theta}_t = 0$ and $\hat{v}_t = \hat{u}_t > 0$ for all $t \geq 1$. Unemployment and vacancies move together along the ray $v = \bar{\theta}u$ at every horizon after impact, and the slope (15) equals $\bar{\theta} > 0$.*

Proof of Proposition 1. Part 1. u_0 is predetermined, so $\hat{u}_0 = 0$. By Lemma 1, $\hat{\theta}_0 = ad_0$. By (12), $\hat{v}_0 = \hat{\theta}_0 + \hat{u}_0 = ad_0$.

Part 2. By Lemma 1, $\hat{\theta}_t = a\rho^t d_0 = -|a|\rho^t d_0$. Substituting into (10) with $d_t = \rho^t d_0$,

$$\hat{u}_{t+1} = \lambda \hat{u}_t + \frac{\bar{f}}{\bar{\tau}} \rho^t d_0 + \omega |a| \rho^t d_0 = \lambda \hat{u}_t + \left(\frac{\bar{f}}{\bar{\tau}} + \omega |a| \right) \rho^t d_0.$$

With $\hat{u}_0 = 0$, iterating gives

$$\begin{aligned}\hat{u}_1 &= \left(\frac{\bar{f}}{\bar{\tau}} + \omega |a| \right) d_0, \\ \hat{u}_2 &= \left(\frac{\bar{f}}{\bar{\tau}} + \omega |a| \right) (\lambda + \rho) d_0,\end{aligned}$$

and in general the expression stated, with $g_t = \sum_{j=0}^{t-1} \lambda^{t-1-j} \rho^j$. Since $\lambda, \rho \geq 0$ and $g_1 = 1$, $g_t > 0$ for $t \geq 1$, and the coefficient in parentheses is positive.

Part 3. By (12), $\hat{v}_1 = \hat{\theta}_1 + \hat{u}_1$. With $\hat{\theta}_1 = a\rho d_0 = -|a|\rho d_0$ and $\hat{u}_1$ from Part 2,

$$\hat{v}_1 = -|a|\rho d_0 + \left(\frac{\bar{f}}{\bar{\tau}} + \omega |a| \right) d_0 = \left[\frac{\bar{f}}{\bar{\tau}} - |a|(\rho - \omega) \right] d_0,$$

which is negative if and only if (14) holds.

Part 4. By (12) and Part 2, for $t \geq 1$,

$$\hat{v}_t = -|a|\rho^t d_0 + \left(\frac{\bar{f}}{\bar{\tau}} + \omega |a| \right) g_t d_0.$$

For $\rho > 0$, divide by $\rho^t d_0 > 0$. Using $g_t/\rho^t = \rho^{-1} \sum_{i=0}^{t-1} (\lambda/\rho)^i$,

$$\hat{v}_t < 0 \iff |a| > \left(\frac{\bar{f}}{\bar{\tau}} + \omega |a| \right) \frac{1}{\rho} \sum_{i=0}^{t-1} \left(\frac{\lambda}{\rho} \right)^i.$$

The right side is strictly increasing in t . So once the inequality fails, it fails at all later dates. This is the first claim.

- If $\rho \leq \lambda$, the sum diverges, so $\hat{v}_t$ eventually turns positive.
- If $\rho > \lambda$, the right side converges from below to $(\bar{f}/\bar{\tau} + \omega |a|)/(\rho - \lambda)$. Then $\hat{v}_t < 0$ for every t if and only if $|a|(\rho - \lambda) \geq \bar{f}/\bar{\tau} + \omega |a|$, that is, $|a|(\rho - \lambda - \omega) \geq \bar{f}/\bar{\tau}$.

For $\rho = 0$, $\hat{v}_t = \hat{u}_t > 0$ for $t \geq 1$.

Part 5. Write $\kappa_u \equiv \bar{f}/\bar{\tau} + \omega |a|$. Since $\tilde{u}_t \approx \bar{u} \hat{u}_t$ and $\tilde{v}_t \approx \bar{v} \hat{v}_t$,

$$\tilde{u}_t = \bar{u} \kappa_u d_0 g_t, \quad \tilde{v}_t = \bar{v} (a\rho^t + \kappa_u g_t) d_0,$$

with $g_0 = 0$. For $\rho \neq \lambda$, $g_t = (\rho^t - \lambda^t)/(\rho - \lambda)$. The case $\rho = \lambda$ follows by continuity. Summing geometric series,

$$\sum_{t \geq 0} \rho^t g_t = \frac{\rho}{(1 - \rho^2)(1 - \rho\lambda)}, \quad \sum_{t \geq 0} g_t^2 = \frac{1 + \rho\lambda}{(1 - \rho^2)(1 - \lambda^2)(1 - \rho\lambda)}.$$

The first uses $\frac{1}{1 - \rho^2} - \frac{1}{1 - \rho\lambda} = \frac{\rho(\rho - \lambda)}{(1 - \rho^2)(1 - \rho\lambda)}$. The second uses

$$(1 - \rho\lambda)(1 - \lambda^2) - 2(1 - \rho^2)(1 - \lambda^2) + (1 - \rho^2)(1 - \rho\lambda) = (\rho - \lambda)^2(1 + \rho\lambda).$$

Therefore

$$\frac{\sum_t \tilde{v}_t \tilde{u}_t}{\sum_t \tilde{u}_t^2} = \frac{\bar{v} \bar{u} \kappa_u d_0^2 (a \sum \rho^t g_t + \kappa_u \sum g_t^2)}{\bar{u}^2 \kappa_u^2 d_0^2 \sum g_t^2} = \bar{\theta} \left[1 + \frac{a}{\kappa_u} \frac{\sum \rho^t g_t}{\sum g_t^2} \right] = \bar{\theta} \left[1 - \frac{|a|}{\kappa_u} \frac{\rho(1 - \lambda^2)}{1 + \rho\lambda} \right],$$

which is (15). It is negative if and only if $|a|\rho(1 - \lambda^2) > \kappa_u(1 + \rho\lambda)$. Substituting $\kappa_u = \bar{f}/\bar{\tau} + \omega|a|$ and rearranging gives (16).

Corollary 1. With $\rho = 0$, $\hat{\theta}_t = a \cdot 0^t = 0$ for $t \geq 1$, so $\hat{v}_t = \hat{u}_t$ by (12). In (15) the correction term has the factor $\rho = 0$, so the slope is $\bar{\theta}$. $\square$

7 Interpretation

Why tightness falls At $\rho = 0$, (9) reads $\eta_L \hat{\theta}_0 = -[1/(r + \bar{\delta}) + 1/(1 - \bar{\delta})]d_0$.

- The vacancy's flow value, and hence the cost of a hire, rises by $1/(r + \bar{\delta})$ in proportion. That is 167 at D1.
- The value of a hire falls only through survival, by $1/(1 - \bar{\delta}) \approx 1$.
- The expected time to fill, $1/q$, must therefore fall sharply. So tightness falls.

The iid case Tightness and vacancies fall only on impact, while unemployment is predetermined. At $t = 1$ the flow value is back to $\bar{K}$ and $\theta_1 = \bar{\theta}$. Unemployment rises, and $v_1 = \bar{\theta}u_1$ rises with it. Entry refills the stock. In the (u, v) plane the economy drops straight down, jumps up and to the right, and returns along the ray $v = \bar{\theta}u$.

At D1, of $\hat{u}_1 = 0.16\%$ per 1% shock:

- 0.04 comes from destruction, $(\bar{f}/\bar{\tau})d_0$;
- 0.12 comes from the date-0 collapse in hiring, $\omega|\hat{\theta}_0|$.

The persistent case Tightness stays low while destruction stays high, so vacancies stay below steady state while unemployment builds up. Once tightness has recovered enough, $v = \theta u$ turns positive and stays positive.

Where $\bar{\delta}$ relative to $\bar{\tau}$ enters Conditions (14) and (16) compare two sensitivities to the same one-point rise in destruction:

- tightness, $|a|$, which is roughly proportional to $1/(r + \bar{\delta})$;
- unemployment, $\bar{f}/\bar{\tau}$.

Every other weight is close across calibrations (Table 1). So the comparison is between $r + \bar{\delta}$ and $\bar{\tau}$.

- At D1, $(r + \bar{\delta})/\bar{\tau} = 0.19$.
- With $\bar{\delta} = \bar{\tau}$ (CK, $s = 0$), $(r + \bar{\delta})/\bar{\tau} = 1.11$. The vacancy dies with the match, so its flow value cannot be the more sensitive of the two.

t	D1			CK-like ($\bar{\delta} = \bar{\tau}$)		
	$\hat{\theta}_t$	$\hat{u}_t$	$\hat{v}_t$	$\hat{\theta}_t$	$\hat{u}_t$	$\hat{v}_t$
0	-0.79	0.00	-0.79	-1.46	0.00	-1.46
1	-0.47	0.17	-0.30	-0.87	0.66	-0.20
2	-0.28	0.19	-0.09	-0.51	0.76	0.25
3	-0.16	0.16	0.00	-0.30	0.66	0.36
4	-0.10	0.13	0.03	-0.18	0.51	0.33

Table 2: Impulse responses to a persistent destruction shock ($\rho = 0.592$, $\kappa = 0$).

	(14)	slope, iid	slope, $\rho = 0.592$	ρ needed for slope < 0	$(r + \bar{\delta})/\bar{\tau}$
D1	holds	0.51	-0.24	$\rho > 0.37$	0.19
CK-like	holds	0.51	0.17	none	1.11

Table 3: Conditions of Proposition 1 and the path slope (15).

8 Numbers

The calibration uses $\kappa = 0$ and $\rho = 0.592$, and responses are per 1% rise in δ_0 , in percent. At D1, $a = -292.0$ (iid: -280.4). At $\bar{\delta} = \bar{\tau}$, $a = -47.2$ (iid: -50.3).

Condition (14) holds in both economies, so the sign at $h = 1$ does not separate them. The path condition (16) does.

9 Remarks

Remark 2. Holding K fixed. If the vacancy cost were a constant flow, rather than the flow value of an asset that δ_t depreciates, the $1/(r + \bar{\delta})$ terms would drop out of (9) and (13). Only the survival effect would remain, and $|a| = 1/\{(1 - \bar{\delta})(\eta_L - B\rho[(1 - \bar{s})\eta_L - \phi\bar{f}])\}$ is small. At our calibration the path slope is then positive in every economy (about 0.49 at $\rho = 0.592$). This is the flow-cost DMP case.

Remark 3. With a per-match cost $\kappa > 0$, the flow-value terms in (9) are scaled by $x_v = (\bar{K}/\bar{q})/(\kappa + \bar{K}/\bar{q})$. At $x_v = 0.5$ the path slopes at $\rho = 0.592$ are -0.23 (D1) and 0.17 ($\bar{\delta} = \bar{\tau}$).

Remark 4. A permanent shock ($\rho \rightarrow 1$) recovers the steady-state comparison. It hides Corollary 1: without persistence, the response is positively sloped after impact.

Remark 5. prop:ds_asymmetry Part 3 in the draft gives “ $\bar{\delta}_e/\bar{\tau}$ small” as a sufficient condition. In the iid free-entry case, entry refills the stock at $t + 1$, and that condition fails for every $\bar{\delta}_e/\bar{\tau}$.